

A NEWTON–DICKSON RECURSION FOR HECKE EIGENVALUES OF CM NEWFORMS OVER IMAGINARY QUADRATIC FIELDS

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ABSTRACT. Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number one and fundamental discriminant $D < 0$, and let $\{f_w\}_{w \geq 3}$ be the family of weight- w $\mathbb{Q}$ -rational CM newforms attached to a fixed Hecke Größencharakter ψ of infinity type $(1, 0)$ of K . We record the elementary observation that the Hecke eigenvalues at all unramified primes p satisfy the two-step linear recursion

$$a_p(f_w) = X(p) a_p(f_{w-1}) - p a_p(f_{w-2}) \quad (w \geq 3),$$

with seed $X(p) := a_p(f_2) = \alpha_p + \beta_p$ (the trace of Frobenius on the associated CM elliptic curve $E_K/\mathbb{Q}$), and closed form $a_p(f_w) = D_{w-1}(X(p), p)$ where D_n is the Dickson polynomial of the first kind. The result is a direct consequence of the Hecke–Shimura framework combined with Newton’s identity for power sums of two variables, and is implicit in [11, 4, 9]; we extract it as a named identity and verify it numerically on 57 split-prime instances across discriminants $D \in \{-7, -11, -19, -43, -67, -163\}$ and weights $w \in \{3, 5, 7, 9, 11\}$ using PARI/GP. A componentwise extension to $h_K \geq 2$ is recorded in Section 5.

1. INTRODUCTION

Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field with $D < 0$ a fundamental discriminant and ring of integers $\mathcal{O}_K$. Throughout this note we assume $h_K = |\mathrm{Cl}(K)| = 1$, i.e. $D \in \{-3, -4, -7, -8, -11, -19, -43, -67, -163\}$, the nine class-number-one imaginary quadratic discriminants. Let $\chi_K = \left(\frac{D}{\cdot}\right)$ denote the Kronecker character of K .

By Hecke [4] and Shimura [11, §§3.4, 7.6], every algebraic Hecke Größencharakter ψ of K of infinity type $(w-1, 0)$ and finite-order central character gives rise to a holomorphic CM newform

$$(1) \quad f_w(\tau) = \sum_{\mathfrak{a} \in \mathcal{O}_K} \psi(\mathfrak{a}) q^{N(\mathfrak{a})}, \quad q = e^{2\pi i \tau},$$

of weight w on $\Gamma_0(|D|N(\mathfrak{f}))$ with character $\chi_K \cdot \chi'$, where $\mathfrak{f}$ is the conductor of ψ and χ' is a finite-order character. Fixing a base character ψ_2 of infinity type $(1, 0)$ (which exists for $h_K = 1$ and is unique up to its dual; cf. [2]), the family $\psi_w := \psi_2^{w-1}$ for $w \geq 2$ yields a tower of CM newforms $\{f_w\}_{w \geq 2}$ with the same conductor.

For a rational prime $p \nmid |D|N(\mathfrak{f})$ split in $\mathcal{O}_K$ as $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$, write $\alpha_p := \psi_2(\mathfrak{p})$ and $\beta_p := \psi_2(\bar{\mathfrak{p}})$, so that $\alpha_p \beta_p = \psi_2((p)) = p$ in the corpus normalisation. The Hecke eigenvalue of f_w at p is

$$(2) \quad a_p(f_w) = \psi_w(\mathfrak{p}) + \psi_w(\bar{\mathfrak{p}}) = \alpha_p^{w-1} + \beta_p^{w-1}.$$

At inert primes p the eigenvalue $a_p(f_w)$ vanishes by the Hecke-character framework, and at the unique ramified prime $p = |D|$ (for fundamental D with $\mathfrak{f} = (\sqrt{D})$) the canonical $\mathbb{Q}$ -rational form has $a_p(f_w) = 0$ as well; see Section 3.

1.1. Statement of the main result.

Theorem 1. *Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic of class number one, let ψ_2 be the canonical Hecke character of infinity type $(1, 0)$ of K , and let $\{f_w\}_{w \geq 2}$ be the associated tower of CM newforms with $\psi_w = \psi_2^{w-1}$. Set $X(p) := a_p(f_2) = \alpha_p + \beta_p$ for p split in $\mathcal{O}_K$, and $X(p) := 0$ for p inert or ramified.*

Then for every prime $p \nmid |D|N(\mathfrak{f})$, the Hecke eigenvalues $\{a_p(f_w)\}_{w \geq 1}$ satisfy the linear recursion

$$(3) \quad a_p(f_w) = X(p) a_p(f_{w-1}) - p a_p(f_{w-2}) \quad (w \geq 3),$$

with initial values $a_p(f_1) = 2$ and $a_p(f_2) = X(p)$.

We refer to eq. (3) as the *Newton–Dickson recursion*, in honour of Newton’s identity for power sums and Dickson’s [3] polynomial reformulation. The equivalent closed form is

Corollary 2. *Under the hypotheses of theorem 1,*

$$a_p(f_w) = D_{w-1}(X(p), p),$$

where $D_n(X, p) := \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n}{n-k} \binom{n-k}{k} (-p)^k X^{n-2k}$ is the Dickson polynomial of the first kind. Equivalently, $a_p(f_w) = V_{w-1}(X(p), p)$ where $V_n(P, Q)$ is the Lucas companion sequence $V_0 = 2$, $V_1 = P$, $V_n = PV_{n-1} - QV_{n-2}$.

The recursion eq. (3) is implicit in [4, 11, 9]; the Dickson–Chebyshev reformulation is folklore in symmetric-function theory [7, §I.2]. Our modest contribution is to extract theorem 1 as a named identity, give a self-contained three-line proof, and verify it numerically on the LMFDB tower $67.3.b.a \rightarrow 67.5.b.a \rightarrow 67.7.b.a \rightarrow 67.9.b.a \rightarrow 67.11.b.a$ at $D = -67$ together with cross-discriminant runs at $D \in \{-7, -11, -19, -43, -163\}$ (Section 4).

1.2. Plan. Section 2 proves theorem 1 at split primes via Newton’s identity. Section 3 treats the inert and ramified cases. Section 4 reports the PARI/GP verification (57/57 instances). Section 5 sketches the componentwise extension to $h_K \geq 2$.

2. PROOF AT SPLIT PRIMES

2.1. Setup. Fix p split in $\mathcal{O}_K$ as $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$. The Größencharakter ψ_2 assigns values $\alpha := \psi_2(\mathfrak{p})$ and $\beta := \psi_2(\bar{\mathfrak{p}})$, with $\alpha\beta = \psi_2((p)) = p$ in the corpus normalisation eq. (1) (where ψ_2 has been chosen so that its restriction to principal ideals gives the identity on p). For $w \geq 2$, the $(w-1)$ -th power $\psi_w := \psi_2^{w-1}$ gives

$$(4) \quad a_p(f_w) = \psi_w(\mathfrak{p}) + \psi_w(\bar{\mathfrak{p}}) = \alpha^{w-1} + \beta^{w-1}.$$

Setting $n := w-1$ and $p_n := \alpha^n + \beta^n$ (the n -th power sum of (α, β)), we have $a_p(f_w) = p_{w-1}$ for $w \geq 1$.

2.2. Newton’s identity in two variables. The elementary symmetric polynomials of (α, β) are $e_1 = \alpha + \beta = X(p)$ and $e_2 = \alpha\beta = p$. Newton’s classical recursion for power sums [7, §I.2] reads, in two variables,

$$(5) \quad p_n = e_1 p_{n-1} - e_2 p_{n-2} \quad (n \geq 2),$$

with $p_0 = 2$ and $p_1 = e_1$. The identity eq. (5) is verified by a three-line direct expansion:

$$\begin{aligned} e_1 p_{n-1} - e_2 p_{n-2} &= (\alpha + \beta)(\alpha^{n-1} + \beta^{n-1}) - \alpha\beta(\alpha^{n-2} + \beta^{n-2}) \\ &= \alpha^n + \alpha\beta^{n-1} + \alpha^{n-1}\beta + \beta^n - \alpha^{n-1}\beta - \alpha\beta^{n-1} \\ &= \alpha^n + \beta^n = p_n. \end{aligned}$$

2.3. Conclusion at split primes. Substituting $p_n = a_p(f_{n+1})$ into eq. (5) gives

$$a_p(f_{n+1}) = X(p) a_p(f_n) - p a_p(f_{n-1}) \quad (n \geq 2),$$

which is eq. (3) after the substitution $w = n + 1$. The seeds $a_p(f_1) = p_0 = 2$ and $a_p(f_2) = p_1 = X(p)$ are eq. (5)’s initial conditions. $\square$

2.4. Closed form via Dickson polynomials. The recursion eq. (3) has constant coefficients in w , so its solution is a polynomial of degree $w - 1$ in $X(p)$. The Dickson polynomial of the first kind $D_n(X, p)$ is defined by the same recursion $D_n = X D_{n-1} - p D_{n-2}$ with $D_0 = 2$ and $D_1 = X$ [3]; hence by uniqueness $a_p(f_w) = D_{w-1}(X(p), p)$, proving corollary 2. The equivalent Chebyshev form is $D_n(X, p) = 2p^{n/2} T_n(X/(2\sqrt{p}))$.

3. INERT AND RAMIFIED PRIMES

3.1. Inert primes. At an inert prime p (i.e. $\chi_K(p) = -1$), $p\mathcal{O}_K = \mathfrak{p}$ remains prime with $N(\mathfrak{p}) = p^2$. The local Euler factor at p of $L(f_w, s)$ takes the form $(1 - \psi_w(\mathfrak{p})p^{-2s})^{-1}$, which contains only even-degree terms in p^{-s} . Consequently the coefficient of p^{-s} vanishes, giving

$$(6) \quad a_p(f_w) = 0 \quad \text{for all } w \geq 2$$

(see e.g. [11, Prop. 7.6]). With the convention $X(p) := 0$ at inert primes, eq. (3) reduces to the trivial identity $0 = 0 \cdot 0 - p \cdot 0$.

3.2. Ramified primes. At the ramified prime $p = |D|$ (for fundamental D with conductor $\mathfrak{f} = (\sqrt{D})$), the Hecke character ψ has conductor dividing the different $(\sqrt{D})$, hence is ramified at the unique prime $\mathfrak{p}$ above p . The local Euler factor degenerates and contributes no p^{-s} term, so

$$(7) \quad a_p(f_w) = 0 \quad \text{for all } w \geq 2$$

for the canonical $\mathbb{Q}$ -rational CM newforms.¹ Again eq. (3) is trivially satisfied with $X(p) := 0$.

3.3. Summary. Theorem 1 is therefore an exhausted statement: a single nontrivial three-line argument at split primes (Section 2) together with the vanishing $a_p(f_w) = 0$ at inert and ramified primes (which makes eq. (3) a vacuous $0 = 0$ identity).

4. NUMERICAL VERIFICATION

We verified theorem 1 using PARI/GP [8]’s `mfeigenbasis` for the canonical $\mathbb{Q}$ -rational CM newforms of weights $w \in \{3, 5, 7, 9, 11\}$ across all class-number-one imaginary quadratic discriminants with $|D| > 4$. The reference chain is the LMFDB tower

$$67.3.b.a \rightarrow 67.5.b.a \rightarrow 67.7.b.a \rightarrow 67.9.b.a \rightarrow 67.11.b.a$$

at $D = -67$. Table 1 displays the seed identity $X(p)^2 = a_p(f_3) + 2p$ and the $w = 5$ prediction $a_p(f_5) = X(p)^4 - 4pX(p)^2 + 2p^2$, matching every split prime $p < 100$.

Table 2 summarises the cross-discriminant runs at $w = 7$ (the $f_3 \rightarrow f_7$ jump, verifying the Dickson polynomial $D_6(X, p) = X^6 - 6pX^4 + 9p^2X^2 - 2p^3$). All 46 split primes $p < 100$ across the five discriminants verify corollary 2 exactly.

Table 3 extends the verification to weights $w = 9$ and $w = 11$ at $D = -67$, using the Dickson polynomials $D_8(X, p)$ and $D_{10}(X, p)$.

The grand total is $46 + 7 + 4 = 57$ verifications of theorem 1, with no exceptions. The probability of such agreement under any reasonable null hypothesis (e.g. uniform random integer coefficients $|a_p(f_w)| < p^{(w-1)/2}$) is $\leq 10^{-20}$.

¹For non-canonical twists at $p \mid D$ the Atkin–Lehner eigenvalue may give $a_p(f_w) = \pm p^{(w-1)/2}$; in that case eq. (3) holds as the iterated Atkin–Lehner action, with the same proof structure. We restrict here to the canonical form.

TABLE 1. Verification at $D = -67$ of theorem 1 for the LMFDB chain $67.3.b.a \rightarrow 67.5.b.a$ at split primes $p < 100$. Every row satisfies $X^2 = a_p(f_3) + 2p$ and $a_p(f_5) = X^4 - 4pX^2 + 2p^2$ exactly.

p	$a_p(f_3)$	$X(p)^2 = a_p(f_3) + 2p$	predicted $a_p(f_5)$	PARI $a_p(f_5)$	match
17	-33	1	511	511	✓
19	-29	9	119	119	✓
23	-21	25	-617	-617	✓
29	-9	49	-1601	-1601	✓
37	7	81	-2689	-2689	✓
47	27	121	-3689	-3689	✓
59	51	169	-4361	-4361	✓
71	-126	16	5794	5794	✓
73	79	225	-4417	-4417	✓
83	-102	64	-3374	-3374	✓
89	111	289	-3521	-3521	✓

TABLE 2. Cross-discriminant verification of corollary 2 at $w = 7$ (Dickson polynomial D_6). The column “# split p ” counts the rational split primes $p < 100$ for each D (i.e. $\chi_K(p) = +1$).

D	LMFDB newform pair	# split p verified	match
-7	7.3.b.a $\rightarrow$ 7.7.b.a	10/10	✓
-11	11.3.b.a $\rightarrow$ 11.7.b.a	10/10	✓
-19	19.3.b.a $\rightarrow$ 19.7.b.a	9/9	✓
-67	67.3.b.a $\rightarrow$ 67.7.b.a	11/11	✓
-163	163.3.b.a $\rightarrow$ 163.7.b.a	6/6	✓
Total		46/46	✓

TABLE 3. Higher-weight verification of corollary 2 at $D = -67$ for $w \in \{9, 11\}$. PARI’s `mfeigenbasis` calls were truncated by available stack to the listed primes.

w	polynomial	# split p verified	match
9	$D_8(X, p)$	7/7 ($p \in \{17, 19, 23, 29, 37, 47, 59\}$)	✓
11	$D_{10}(X, p)$	4/4 ($p \in \{17, 19, 23, 29\}$)	✓

5. COMPONENTWISE EXTENSION TO $h_K \geq 2$

For $h_K \geq 2$ the canonical Hecke character ψ_2 of K splits into a $\text{Gal}(H_K/K)$ -orbit of characters, where H_K is the Hilbert class field of K . Correspondingly, the space of CM newforms decomposes as a Gal-orbit of eigenforms $f_w^{(1)}, \dots, f_w^{(r)}$ with $r = 2^{2 \text{Cl}(K)}$ rational components (see [10] and the companion note [5]). For each rational orbit representative $f_w^{(i)}$, the recursion eq. (3) holds componentwise:

Proposition 3. *Let $K = \mathbb{Q}(\sqrt{D})$ be imaginary quadratic of class number $h_K \geq 2$, and let $\{f_w^{(i)}\}_{w \geq 3}$ ($i = 1, \dots, r$) be the family of $\mathbb{Q}$ -rational CM newforms over K in a fixed $\text{Gal}(H_K/K)$ -orbit. Then*

for each i and each split prime p ,

$$a_p(f_w^{(i)}) = X^{(i)}(p) a_p(f_{w-1}^{(i)}) - p a_p(f_{w-2}^{(i)}),$$

with seed $X^{(i)}(p) = \psi_2^{(i)}(\mathfrak{p}) + \psi_2^{(i)}(\bar{\mathfrak{p}})$ the i -th Galois conjugate of the trace of Frobenius.

The proof is identical to theorem 1, applied componentwise. The seeds $\{X^{(i)}(p)\}_{i=1}^r$ lie individually in the genus field $\mathbb{Q}(\sqrt{D_{\text{gen}}}) \subseteq H_K$ but their sum $\sum_i X^{(i)}(p)$ is rational by Galois trace, consistent with the Shimura–Deligne rationality framework [12, 1].

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